Cell biology

Structure & observation

1 Cells & specialisation

Animal: nucleus → DNA/control; cytoplasm → reactions; membrane → entry/exit.

Mitochondria: aerobic respiration.
Ribosomes: make proteins.

Plant/algal extras: cellulose wall; permanent vacuole. Photosynthetic cells have chloroplasts.

Bacteria: smaller; no nucleus; DNA loop; some plasmids.

Sperm: tail, mitochondria, acrosome enzymes.
Nerve: long fibre, branches.

Muscle: contractile proteins, mitochondria.
Root hair: large absorption area.

Xylem: hollow, lignified; water/minerals.
Phloem: sugars; companion cells.

2 Microscopy

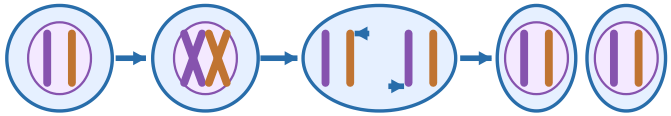
Resolution: distinguish nearby points. Electron microscopes give more detail and magnification.

Magnification = image ÷ actual size. Same units first: mm → µm ×1,000.

Practical: thin stained sample, angled coverslip; low power first; focus, draw, label and add a scale.

Making new cells

3 Division & stem cells



Example shown: **two chromosomes.**

Grow → copy DNA → mitosis → divide.

Copies separate; nuclei, then cytoplasm/membrane divide. **Two identical cells; same chromosome number.**

For growth, development, repair and replacement.

Differentiation: becoming specialised; animals mainly early, plants throughout life.

Embryonic: most human cell types. **Bone marrow:** limited types. **Meristems:** any plant type.

Uses: tissue repair; plant clones. Weigh benefits, infection risk and ethical objections.

4 Culturing microorganisms

Binary fission: repeated doubling. Population = start × 2^n; n = time ÷ division time.

Asepsis: sterilise, cool loop; tape edges, invert; school cultures **≤25°C.**

Compare antimicrobial discs; solvent control; repeat. **Zone area = πr^2;** r = diameter ÷ 2.

Movement & exchange

5 Diffusion & exchange

Diffusion: net particles high → low; no respiration energy. Faster: steeper gradient, hotter, larger area.

Large organisms: **lower SA:V** → exchange/transport systems. Cube: SA = $6a^2$; V = a^3.

Alveoli, villi, gills: large area, thin barrier, blood supply; gas exchange needs ventilation.

6 Osmosis & active transport

Osmosis: water, dilute → concentrated, through a partially permeable membrane.

Water gain: plant turgid; animal swells. Loss: plant flaccid; animal shrinks.

Practical: equal plant samples; weigh, soak, blot, reweigh. **% change = (final − initial) ÷ initial × 100.**

Active transport: low → high; respiration energy. Root minerals; gut sugars.